

Kinematic and Dynamic Modeling of a Generalized Tractor-Trailer System with Steering Actuation

Marcel Moll* Timo Oksanen*

* Chair of Agrimechatronics, Munich Institute of Robotics and Machine Intelligence (MIRMI), Technical University of Munich, Germany (e-mail: {marcel.moll; timo.oksanen}@tum.de)

Abstract: The study of vehicle behavior for the development of guidance algorithms has been widely explored. In agriculture, tractor guidance systems are now standard features in premium machinery. The next step to further improve precision is to achieve centimeter-level guidance for connected implements, especially those equipped with additional steering capabilities. Compared to the tractor-only scenario, the complexity of model design increases, and there is no established consensus among manufacturers on the preferred modeling approach. To facilitate interoperability between implement and tractor manufacturers, a generalized approach is required and has been previously proposed. This work presents an enhanced model that features an abstracted control interface for actuated towed implements. This interface is designed to be extendable to hitched implements and can account for known tire side-slip. For more detailed simulations, a dynamic version of the model is derived, allowing for the use of advanced tire and force models. The paper includes a systematic derivation of both kinematic and dynamic models, as well as first simulation results.

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1. INTRODUCTION

Guidance systems based on Global Navigation Satellite Systems (GNSS) for tractors have become a standard feature in premium machinery. They are also increasingly available as retrofit options for cost-effective or older tractors. These systems utilize Real Time Kinematic or Precise Point Positioning through GNSS and claim tracking accuracies at the centimeter level. For precision agriculture, the primary focus is however not on the tractor's positioning but rather on the implement. Especially long trailed implements tend to drift due to external forces induced by slopes, unbalanced draft forces or slippery ground conditions. While purely trailed implements can only be corrected by adjusting the tractor's guidance, actuated implements with steering capabilities have been proposed. These systems are significantly more complex in terms of modeling and control and are often not produced by the same brand as the guidance system. In addition, steering actuators may be part of the implement concept and required to properly fulfill the agronomic task.

Figure 1 shows a typical scene on a field: A potato harvester is offloading harvested and processed crops into a tipper trailer. Below, the representative models usable for guidance applications are sketched. The most relevant features are indicated by circled numbers.

While the tipper trailer lacks active steering capabilities, the potato harvester is equipped with both a hydraulically steerable drawbar and rear axle.

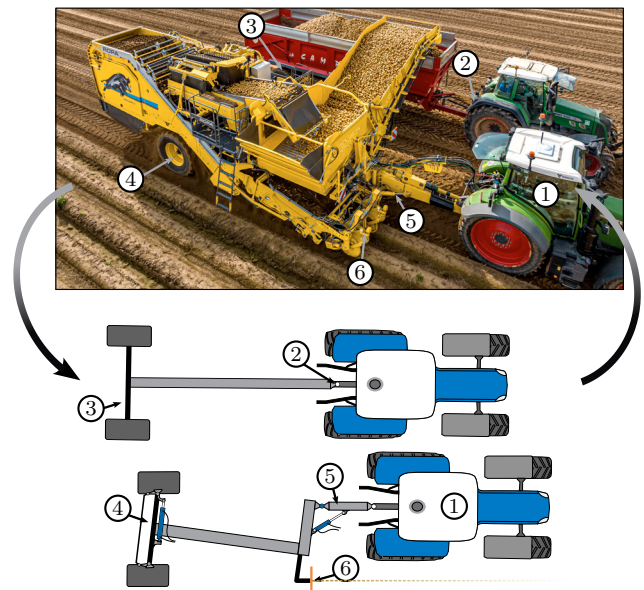


Fig. 1. Conversion from real-world tractor-trailer combinations to the simplified models for guidance applications (Imaged based on AGTRACKS.TG (2023))

- | | |
|---------------------|--------------------------|
| ① Tractor | ② Drawbar coupler |
| ③ Fixed rear axle | ④ Steerable rear axle |
| ⑤ Steerable drawbar | ⑥ Guidance control point |

Both actuators ④ and ⑤ are used in conjunction to shift the potato pick-up ⑥ laterally with regards to the tractor and align the intake with the ridges.

1.1 Current State of Research

Given the complexity of precisely controlling these machines, numerous studies have focused on their modeling and control.

Karkee and Steward (2008) derived a kinematic and a dynamic model for a tractor with trailer. Based on eigenvalue analysis of both models, the system behavior was found to be approximated equally well for low-speed applications. In Backman et al. (2009), a guidance system for a tractor-trailer system with a steerable drawbar was developed. A specialized model was derived that incorporated a laser scanner to measure and control the relative distance to a previously made furrow. The nonlinear model was linearized for the Model Predictive Controller and was only updated when the heading changed by 45° , which negatively impacted tracking performance.

Murillo, M. et al. (2022) derived a kinematic model for predictive tractor-trailer guidance, in which the tractor is capable of both articulated steering and front-wheel steering.

In Werner et al. (2013), a tractor-trailer system with a steerable drawbar and rear axle was controlled using a Linear Quadratic Regulator based on a kinematic model. In addition to the kinematic model, a dynamic representation of the same tractor-trailer system was previously derived in Werner et al. (2012). Later, Werner (2015) indicated that this dynamic model required too many parameters and was too complex to be used as a reference for a guidance controller. Instead, extensive simulations of the vehicle behavior were conducted using the dynamic model with variations in tire properties and masses.

Augmenting the kinematic model of a tractor-trailer system with slip parameters was conducted in Cariou et al. (2010). A dedicated observer provided the slip parameters as inputs to the kinematic model. Experimental results demonstrated that the control system for the non-actuated trailer significantly improved tracking of curved paths and mitigated the effects of sloped terrain.

So far, the studies presented are considering a single system, either unsteered or with a known set of actuators and a defined geometry. This approach is sufficient if a dedicated controller shall guide the particular implement and has full access to control interfaces of both the tractor and implement. In practices, this is hardly the case and may lead to a number of stand-alone solutions that are incompatible with other machines.

For a tractor-trailer combination, which uses multiple unsteered trailers, Bulgakov et al. (2019) presented a more generalized approach for retrieving a dynamic model. To avoid rederiving the equations of motion when adding or removing a new unsteered trailer, a parametric formalism based on Hamel's equations of motion has been exploited. Simulations are provided for a large seeder combination with four individual trailers, incorporating draft forces obtained from experiments with a real machine.

With the goal of standardizing a single implement guidance model applicable to the most common actuated tractor-trailer combinations, Oksanen and Backman (2016)

presented a generalized kinematic model. A more detailed approach tailored for the ISO 11783 standardization committee can be found in Oksanen and Backman (2015). However, the standardization committee has not yet prioritized the standardization of implement guidance features across manufacturers.

1.2 Scope of this Work

The goal of this paper is to extend the concept of a generalized implement model, building on the work presented in Oksanen and Backman (2015), with regards to the growing demand for higher precision in planting, fertilizing, and spraying with large towed implements. The objective is to introduce an improved control interface that supports the use of multiple actuators, accompanied by a detailed derivation in the first part of the paper. Additionally, the paper aims to extend the method for obtaining the kinematic model by incorporating tire slip on all axles. Subsequently, a dynamic model is developed to enhance modeling fidelity, offering optional interfaces for accurate tire models, draft forces, and gravitational influences.

2. KINEMATIC AND DYNAMIC MODELING

2.1 Model applicability

The kinematic and dynamic models are designed to cover the majority of actively steered implements commonly used in agricultural fieldwork. Due to the limitations in generalizing models, not all implements can be practically incorporated into a single representation. Figure 2 illustrates the applicable implement types, as well as those commonly available on the market. In analogy to Oksanen and Backman (2015), the control interfaces of *sideshift* and *body-offset angle* are applicable to most relevant implement types with steering actuators and would only require different kinematic or dynamic models. For the remainder of this work, only the green-colored implement types in fig. 2 are considered.

2.2 Abstracted Tractor-Trailer Model

The sketch of a tractor with an attached trailer is illustrated in fig. 3. A bicycle representation is used, which lumps the kinematically redundant wheel pairs into a single combined wheel and is sufficient for deriving the planar motion of the vehicle. In fig. 3, the tractor is characterized mainly by two parameters: The mounting offset for the trailer connector b , which is defined relative to the vehicle turning-point, and the curvature κ . The curvature describes the arc, tangential to the vehicle forward-axis at the turning-point, on which the vehicle is moving. For a front-steering vehicle, the turning-point coincides with the rear axle center. With a given steering angle $\delta_{t,f}$ and wheelbase l_{wb} , the curvature in this case can be calculated by $\kappa = 1/R = \tan(\delta_{t,f})/l_{wb}$. The ISO 11783-7 (2022) adopts the curvature as generalized steering command, making the controller independent of the actual steering geometry. The tractor has a body-fixed coordinate system, attached to the rear axle center and a corresponding heading angle ψ_t . The gearbox speed v_g defines the velocity of the tractor at the rear axle $\mathbf{v}_{t,r} = [v_g \ 0]^T$ in tractor coordinates.

As the next step, the holonomic constraint of the wheel will be considered. Assuming pure rolling without side-slip, the lateral velocity at the wheel must be zero

$$\mathbf{v}_{t,f}^T \mathbf{R}(\psi_t + \delta_{t,f}) \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0. \quad (4)$$

Note that it is not required to restate the constraint at the rear axle, since the definition of the velocity $\mathbf{v}_{t,r}$ already incorporates the zero-slip constraint. Substituting (3) in (4) yields a single equation, where the only unknown is the heading rate $\dot{\psi}_t$. After some algebraic simplifications, the differential equation for the heading of the front-steered vehicle is

$$\dot{\psi}_t = \frac{v_g \tan \delta_{t,f}}{l_{wb}} = \kappa v_g. \quad (5)$$

In the kinematic modeling approach, the tractor is completely independent of the trailer, as there are no forces, inertia or friction considered. The attached trailer in fig. 3 on the other hand requires the tractor or another reference body, which sets the velocity at the coupling. Still, the same derivation procedure applies.

Prior to stating the distance vector $\mathbf{r}_{i,ar}$ from the tractor rear axle towards the trailer wheel, a few geometric relations are evaluated. The sideshift ΔS can be divided into two parts

$$\Delta S = \Delta S_1 + \Delta S_2, \quad (6)$$

where ΔS_1 stems from the rotation of the main implement body and is given by

$$\begin{aligned} \Delta S_1 &= [0 \ 1] \mathbf{R}(\Delta\theta) \begin{bmatrix} l_1 \\ -l_3 \end{bmatrix} \\ &= l_1 \sin(\Delta\theta) - l_3 \cos(\Delta\theta). \end{aligned} \quad (7)$$

Since ΔS is an input (or control) parameter of the model, the side-shift due to superimposed drawbar joint angle and $\Delta\theta$ is

$$\Delta S_2 = \Delta S - l_1 \sin(\Delta\theta) + l_3 \cos(\Delta\theta). \quad (8)$$

A similar concept is applied for the vertical components L_1 and L_2 , respectively:

$$\begin{aligned} L_1 &= [1 \ 0] \mathbf{R}(\Delta\theta) \begin{bmatrix} l_1 \\ -l_3 \end{bmatrix} \\ &= l_1 \cos(\Delta\theta) + l_3 \sin(\Delta\theta) \end{aligned} \quad (9)$$

and

$$L_2 = \sqrt{l_2^2 - \Delta S_2^2}. \quad (10)$$

The square-root in (10) introduces an input constraint for the maximum permissible sideshift ΔS . It is depending on the choice of $\Delta\theta$ and the trailer dimensions, which can be verified with the help of fig. 3.

Collecting the auxiliary measurements and adding them with regards to the appropriate coordinate system finally yields the distance vector

$$\mathbf{r}_{i,ar} = \mathbf{R}(\psi_t) \begin{bmatrix} -b \\ 0 \end{bmatrix} + \mathbf{R}(\psi_i) \begin{bmatrix} -(L_2 + L_1) \\ \Delta S \end{bmatrix}, \quad (11)$$

which connects the axles of tractor and implement. Formulating the holonomic constraint at the rear axle takes the same form as (4) and reads

$$\mathbf{v}_{i,r}^T \mathbf{R}(\psi_i) \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0. \quad (12)$$

Solving for the heading rate of the trailed implement results in

$$\dot{\psi}_i = \frac{\Delta\dot{S} + v_g \sin(\psi_t - \psi_i) - b \kappa v_g \cos(\psi_t - \psi_i)}{L_1 + L_2}, \quad (13)$$

where the intermediate dimensions L_1 and L_2 are calculated according to (9) and (10), respectively.

Comparing the equations $\dot{\psi}_t$ and $\dot{\psi}_i$ from (5) and (13), they both have the form of a horizontal velocity being divided by the vertical lever-arm towards the axle-center. Further breaking down (13), the horizontal velocity in the nominator can be separated in a joint velocity $v_{b,t}^y$ stemming from the tractor and a component $v_{b,i}^y$ due to the implement movement:

$$\dot{\psi}_i = \frac{v_{b,t}^y + v_{b,i}^y}{L}, \text{ where} \quad (14)$$

$$L = L_1 + L_2, \quad v_{b,i} = \Delta\dot{S} \text{ and}$$

$$v_{b,t}^y = v_g \sin(\psi_t - \psi_i) - b \kappa v_g \cos(\psi_t - \psi_i).$$

For simulating the position of the implement rear axle's on the tractor's guidance controller, it is thus sufficient to only know the initial values of the side-shift ΔS and the combined implement length L and obtain the rates $\Delta\dot{S}$ and $\dot{L}$ regularly, while implement-steering actuators are active. As stated in Oksanen and Backman (2015), the side-shift cannot be used directly as error-variable to minimize a cross-track error to the tractor's path in a PID-like controller. If the tractor is thus intending to precisely control the implement, the simplification is not sufficient anymore and detailed knowledge of the implement kinematics are required, which are available in (13).

A common model for defining the states of the combined tractor-trailer, suited for the guidance task, is

$$\begin{bmatrix} \dot{x}_t^g \\ \dot{y}_t^g \\ \dot{\psi}_t \\ \dot{\psi}_i \end{bmatrix} = \begin{bmatrix} v_g \cos(\psi_t) \\ v_g \sin(\psi_t) \\ \kappa v_g \\ (13) \end{bmatrix}. \quad (15)$$

The tractor's position, given by $\{x_t^g, y_t^g\}$, is expressed in the global coordinate frame. Additional positions or velocities can be calculated using (1), (3) and (11), but these calculations are optional.

A common improvement is the addition of slip parameters, which can be incorporated for each tire in the system. Sliding is modeled laterally to the tire axis using a tire slip angle α , defined counterclockwise, as shown in fig. 4. The

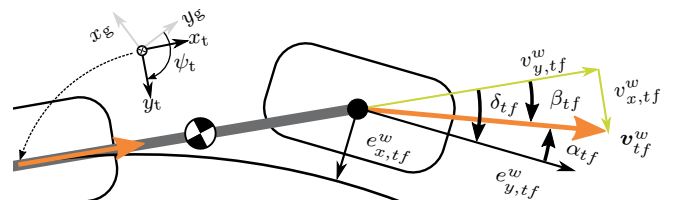


Fig. 4. Slip angle at the tractor front-axle

side slip angle β , measured relative to the vehicle's forward axis, includes the steering angle δ and can be calculated from the wheel's velocity vector $\mathbf{v}^w$. The inclusion to the derivation of the kinematic model above is straight forward: All holonomic constraints are augmented by a respective slip parameter. For the example of the tractor front-axle in (4), this reads

$$\mathbf{v}_{t,f}^T \mathbf{R}(\psi_t + \delta_{t,f} + \alpha_{t,f}) \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0. \quad (16)$$

As mentioned earlier, the slip augmentation of the kinematic model requires an external routine for the estimation or measurement of slip, which is crucial for the accuracy of the model. Due to space restrictions, the extended version of (15) is not printed, but can be derived with the same method as above.

2.4 Dynamic model

If a more consistent slip treatment or in general a higher fidelity model is sought, a dynamic representation may be chosen. The procedure to obtain the dynamic equations of motion for the tractor-trailer combination in fig. 1 is making use of the Lagrangian equations of the second kind and follows the educative examples in Genta and Genta (2017) with adaptations for general external forces by Werner et al. (2012). Therefore, the following steps are performed:

- (1) Definition of generalized coordinates $\mathbf{q}$
- (2) Determination of components required to express the total energy of the system with the Lagrangian function $L = T - V$
- (3) Definition of generalized forces $\mathbf{Q}$
- (4) Evaluation of $\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_j} - \frac{\partial L}{\partial q_j} = Q_j$ to obtain equations of motion
- (5) Reformulating the equations of motion to a linear implicit form
- (6) Definition of actuator and tire models

There are $j = 4$ generalized coordinates required for the system in fig. 1, which are chosen as $\mathbf{q} = [x_t^g \ y_t^g \ \psi_t \ \psi_1]^T$. In the next step, the total energy of the system has to be expressed based on the kinetic energy T and the potential energy V . The kinetic energy is

$$T = \frac{1}{2} [I_t \psi_t^2 + m_t \|\mathbf{v}_t^g\|^2 + I_i \psi_i^2 + m_i \|\mathbf{v}_i^g\|^2] . \quad (17)$$

Herein, the velocities of the centers of mass in the global frame, $\mathbf{v}_t^g$ and $\mathbf{v}_i^g$, were used in combination with the masses and moments of inertia from fig. 1. There is no potential energy V in the system, such that $L = T$.

In contrast to the kinematic model, $\mathbf{v}_i^g$ is calculated without considering the influences of $\Delta\theta$ and is based on the distance vector

$$\mathbf{r}_{i,c}^g = \mathbf{r}_{t,c}^g + \mathbf{R}(\psi_t) \begin{bmatrix} -l_{c,t} - b \\ 0 \end{bmatrix} + \mathbf{R}(\psi_i) \begin{bmatrix} -L_1 - L_2 + l_{i,c} \\ \Delta S \end{bmatrix} . \quad (18)$$

The generalized forces

$$Q_j = \sum_{k=1}^N \mathbf{P}_k \frac{\partial \mathbf{r}_k}{\partial q_j} = \sum_{k=t,i} \begin{bmatrix} F_{k,x}^g \\ F_{k,y}^g \\ M_{k,z}^g \end{bmatrix} \frac{\partial [x_k^g \ y_k^g \ \psi_k^g]}{\partial q_j} \quad (19)$$

are composed of the forces F and moments M , which are acting on the masses of the rigid bodies.

As shown by Genta and Genta (2017), the global entities for forces and velocities can be replaced with their more convenient vehicle-frame counterparts

$$\mathbf{v}_t^g = \mathbf{R}(\psi_t) \mathbf{v}_t, \quad \mathbf{F}_t^g = \mathbf{R}(\psi_t) \mathbf{F}_t, \quad \mathbf{F}_i^g = \mathbf{R}(\psi_i) \mathbf{F}_i \quad (20)$$

before evaluating the time derivative in the Lagrangian equations of motion. Without further detailing the external forces, the equations of motion can already be obtained from

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_j} - \frac{\partial L}{\partial q_j} = Q_j . \quad (21)$$

While the substitution for vehicle-fixed variables in (20) simplifies the use of known relations, it also causes the first two differential equations to resemble an unfavorable local acceleration. To address this, the first two equations of (21) are multiplied by the reverse transformation $\mathbf{R}(\psi_t)^T$.

Although the four equations from (21) are sufficient for simulating the dynamic behavior of the system, a transformation to a linear implicit form

$$\mathbf{M}(\mathbf{y}) \dot{\mathbf{y}} = \mathbf{f}(\mathbf{y}) \quad (22)$$

as done in Werner (2015) is performed. In (22), the mass matrix $\mathbf{M}$ is introduced, which depends on the states $\mathbf{y}$ and significantly reduces the expression length of the right-hand side function $\mathbf{f}(\mathbf{y})$. However, the overall length of the expression makes it impractical to print the symbolically derived results here.

So far, neither slip nor steering angles have been explicitly considered in the dynamic model. The following tire model interacts with the system through external forces and moments. In the literature, numerous tire models are available, differing in modeling approach, required data, and computational effort. Since the overall dynamic model is kept compact, the linear model

$$F_j^k = C_{j,\alpha} \alpha_j \quad (23)$$

is selected. This model calculates the lateral force F_j^k induced by the j -th tire using the cornering stiffness $C_{j,\alpha}$ and the respective slip angle α_j . The slip angle can be calculated based on the velocity vector $\mathbf{v}_j^w$ and the steering angle α_j , as shown in fig. 4. If modeling the transient behavior of the tire is desired, a relaxation length can be introduced using a first-order delay, as discussed by Schlotter (2005):

$$\dot{\alpha}_j = (\alpha_{j,d} - \alpha_{j,r}) \frac{v_{j,x}^w}{\sigma_{j,\alpha}} . \quad (24)$$

Then, the realized slip angle $\alpha_{j,r}$ will converge towards the calculated $\alpha_{j,d}$ after a defined traveled distance, which is set via $\sigma_{j,\alpha}$.

Similarly, actuators can be modeled with either first or second order delays, as demonstrated for both kinematic and dynamic models in Karkee and Steward (2008); Oksanen and Backman (2015); Werner et al. (2013). This enables the model to account for the typical response of a closed-loop controlled actuator.

3. EVALUATION

3.1 Simulation Comparison

A brief comparison of the kinematic model, with and without slip, and the dynamic model was conducted in Matlab. The same model parameters were used when applicable, based on the measurements and estimations by Werner (2015). In the kinematic model with slip, the parameters α_j were calculated using a basic linear relation with the vehicle's heading rate.

The integration of the kinematic model is performed with Matlab's explicit Runge-Kutta (2,3) implementation *ode23*, whereas the dynamic model is handled using the *ode15s* solver for moderately stiff problems with non-constant mass matrices. Internal steps of the variable-step solvers are not evaluated, such that only the computation

time required for a 100 ms step is listed in table 1. The

Table 1. Evaluation of the computation time per 100 ms time step

Model	Solver	Average	Min	Max
① Kinematic w. slip	ode23	0.325 ms	0.308 ms	0.559 ms
② Kinematic no slip	ode23	0.324 ms	0.309 ms	0.468 ms
③ Dynamic	ode15s	1.408 ms	0.642 ms	2.273 ms

timing tests are averaged over 100 simulation runs, performed on an AMD Ryzen 5 7530U with 16 GB RAM. All computations in table 1 are well within the 100 ms simulation time step. While there is a significant increase in computation time from the kinematic to the dynamic model, the addition of slip parameters has no notable effect on the overall computation time.

The simulated trajectories of the tractor and trailed implement are shown in fig. 5. As control input, a sinu-

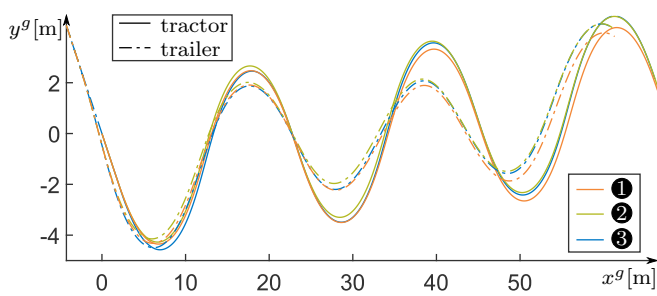


Fig. 5. Comparison of simulation results with sinusoidal actuator inputs for different models

soidal change in the tractor steering angle and implement sideshift is commanded. While all models closely follow a common path, slight variations are observable. Initially, the slip-incorporating kinematic model tracks the dynamic model well but begins to diverge after 35 m. The results suggest that the approximated slip model cannot fully substitute the dynamic model under the given system parameters and reference signals. However, it is important to note that assessing the overall accuracy based on fig. 5 is not possible, as there is no definitive ground truth for comparison.

4. CONCLUSION

This work has focused on developing an abstracted model for trailed implements with steering actuators, intended for precision implement-guidance applications in agriculture. The kinematic and dynamic models, derived step-by-step, were briefly tested through simulations and demonstrated promising execution times. A more in-depth analysis of the models, ideally involving real machines under various operating conditions, remains to be conducted. Open research questions also persist regarding the estimation of slip parameters and the integration of the model into closed-loop control systems. These contributions are intended to enhance the applicability of the generalized model, enabling manufacturers of guidance systems, implements, and tractors to achieve precision implement guidance across multi-brand use cases.

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REFERENCES

- AGTRACKS_TG (2023). Keiler 2 rk22 l überladerbunker 0125. URL <https://www.ropa-maschinenbau.de/mediathek/default/kartoffeltechnik/fotos/>.
- Backman, J., Oksanen, T., and Visala, A. (2009). Parallel guidance system for tractor-trailer system with active joint. *Precision agriculture*, 615–622.
- Bulgakov, V., Ivanovs, S., Adamchuk, V., and Antoshchenkov, R. (2019). Investigations of the dynamics of a four-element machine-and-tractor aggregate. *Acta Technologica Agriculturae*, 22(4), 146–151. doi:10.2478/ata-2019-0026.
- Cariou, C., Lenain, R., Thuilot, B., and Martinet, P. (2010). Path following of a vehicle-trailer system in presence of sliding: Application to automatic guidance of a towed agricultural implement. In *2010 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 4976–4981. doi:10.1109/IROS.2010.5652673.
- Genta, G. and Genta, A. (2017). *Road vehicle dynamics: Fundamentals of modeling and simulation*, volume 88 of *Series on advances in mathematics for applied sciences*. World Scientific, New Jersey.
- ISO 11783-7 (2022). Tractors and machinery for agriculture and forestry — part 7: Implement messages application layer.
- Karkee, M. and Steward, B.L. (2008). Open and closed loop system characteristics of a tractor and an implement dynamic model. In *2008 Providence, Rhode Island, June 29 - July 2, 2008*, volume 10. American Society of Agricultural and Biological Engineers, St. Joseph, MI. doi:10.13031/2013.25208.
- Murillo, M., Sánchez, G., Deniz, N., Genzelis, L., and Giovanini, L. (2022). Improving path-tracking performance of an articulated tractor-trailer system using a non-linear kinematic model. *Computers and Electronics in Agriculture*, 196, 106826. doi:10.1016/j.compag.2022.106826.
- Oksanen, T. and Backman, J. (2015). Standardization proposal on implement guidance for ISO 11783 compatible tractor-implement systems.
- Oksanen, T. and Backman, J. (2016). Implement guidance model for ISO 11783 standard. *IFAC-PapersOnLine*, 49(16), 33–38. doi:10.1016/j.ifacol.2016.10.007.
- Schlotter, V. (2005). *Einfluss dynamischer Radlastschwankungen und Schräglaufwinkeländerungen auf die horizontale Kraftübertragung von Ackerschlepperreifen*. Doctoral thesis, Universität Stuttgart, Hohenheim.
- Werner, R. (2015). *Centimeter-Level Accuracy Path Tracking Control of Tractors and Actively Steered Implements*. Doctoral thesis, Technische Universität Kaiserslautern.
- Werner, R., Kormann, G., and Mueller, S. (2012). Dynamic modeling and path tracking control for a farm tractor towing an implement with steerable wheels and steerable drawbar. In *Commercial Vehicle Technology Symposium*, volume 2, 241–250. Shaker Verlag, Kaiserslautern, Germany.
- Werner, R., Kormann, G., and Mueller, S. (2013). Systematic model based path tracking control of actively steered implements in simulation and experiment. *IFAC Proceedings Volumes*, 46(18), 85–90. doi:10.3182/20130828-2-SF-3019.00015.